TIRust benchmark summary

2026-11-06 · TIRust `a0c4884cc5` · AMD EPYC 9754 128-Core Processor

Ratios are **TIRust/reference**, each measured in the same capture as paired block medians with a 99% bootstrap interval (below one: TIRust faster; green faster, red slower). No ratio is reconstructed across captures. TIRust is built in release mode (thin LTO, one codegen unit) with `-C target-feature=+fma` (bit-identical outputs; huashengshao/TIRust#5); IREGI and COLLIER are built with gfortran `-O3`, baseline x86-64, and NinjaRS with its own release profile. Unmeasured, skipped and failed cases are kept and named. This document keeps the summary tables only; the captures, per-point accuracy deltas and the full reports are archived at https://github.com/huashengshao/TIRust/releases/tag/tirust-bench-20261106.

1 Single integrals

Table 1: Single phase-space point: TIRust (default path, automatic precision rescue) against original IREGI, COLLIER and NinjaRS (contraction with a dense random numerator, against NinjaRS's integrand reduction of it)

Integral	TIRust µs	vs IREGI	vs COLLIER	vs NinjaRS
N=1, R=0	0.36	×**0.47** [0.46, 0.47] 0.78 µs	×**0.93** [0.92, 0.94] 0.39 µs	×**1.42** [1.35, 1.45] 0.25 µs
N=1, R=6	2.96	×**0.13** [0.13, 0.13] 23.4 µs	×**0.63** [0.63, 0.64] 4.66 µs	not measured
N=2, R=1	0.87	×**0.28** [0.27, 0.28] 3.14 µs	×**0.77** [0.77, 0.78] 1.12 µs	×**0.84** [0.81, 0.85] 1.08 µs
N=2, R=6	4.57	×**0.012** [0.012, 0.012] 386 µs	×**0.53** [0.53, 0.54] 8.55 µs	not measured
N=3, R=2	3.76	×**0.15** [0.15, 0.15] 24.8 µs	×**0.45** [0.45, 0.46] 8.27 µs	×**0.082** [0.080, 0.083] 46.2 µs
N=3, R=3	4.36	×**0.064** [0.063, 0.066] 67.2 µs	×**0.41** [0.40, 0.41] 10.7 µs	×**0.061** [0.060, 0.061] 72.9 µs
N=3, R=6	8.55	×**0.0032** [0.0031, 0.0032] 2702 µs	×**0.17** [0.17, 0.17] 50.6 µs	not measured
N=4, R=4	32.5	×**0.061** [0.060, 0.062] 533 µs	×**0.65** [0.64, 0.65] 50.4 µs	×**0.12** [0.12, 0.13] 262 µs
N=4, R=6	49.4	×**0.0040** [0.0039, 0.0040] 12484 µs	×**0.19** [0.18, 0.19] 265 µs	×**0.034** [0.033, 0.034] 1456 µs
N=5, R=5	116	×**0.013** [0.013, 0.013] 8724 µs	×**0.39** [0.38, 0.39] 300 µs	×**0.16** [0.16, 0.16] 730 µs
N=5, R=6	154	×**0.0033** [0.0033, 0.0033] 46123 µs	×**0.14** [0.13, 0.15] 1069 µs	×**0.087** [0.085, 0.087] 1750 µs
N=6, R=6	430	×**0.0021** [0.0021, 0.0022] 202304 µs	×**0.22** [0.22, 0.22] 1952 µs	×**0.32** [0.31, 0.33] 1339 µs
N=7, R=7	2051	×**0.00055** [0.00054, 0.00056] 3711318 µs	×**0.40** [0.40, 0.41] 5079 µs	×**0.59** [0.59, 0.60] 3438 µs
N=1, R=6 (dot2)	3.51	×**0.10** [0.10, 0.10] 34.6 µs	not measured	not measured
N=2, R=2 (dot2)	2.06	×**0.075** [0.075, 0.076] 27.4 µs	not measured	not measured
N=3, R=4 (dot2)	9.26	×**0.017** [0.017, 0.017] 540 µs	not measured	not measured
N=4, R=6 (dot2)	38.7	×**0.0026** [0.0026, 0.0026] 14952 µs	not measured	not measured
N=4, R=6 (dot3)	62.8	×**0.0034** [0.0034, 0.0035] 18212 µs	not measured	not measured
N=3, R=2 (h1)	4.10	×**0.052** [0.052, 0.053] 78.3 µs	not measured	not measured

Table 2: Batch of 32 points, time per point: TIRust (default path, automatic precision rescue) against original IREGI, COLLIER and NinjaRS (contraction with a dense random numerator, against NinjaRS's integrand reduction of it)

Integral	TIRust µs	vs IREGI	vs COLLIER	vs NinjaRS
N=1, R=0	0.17	×**0.22** [0.22, 0.22] 0.79 µs	×**0.42** [0.41, 0.42] 0.42 µs	×**1.47** [1.43, 1.49] 0.12 µs
N=1, R=6	2.71	×**0.12** [0.11, 0.12] 23.4 µs	×**0.57** [0.57, 0.58] 4.75 µs	not measured
N=2, R=1	0.62	×**0.20** [0.20, 0.21] 3.04 µs	×**0.55** [0.55, 0.55] 1.13 µs	×**1.07** [1.06, 1.08] 0.60 µs
N=2, R=6	4.15	×**0.011** [0.0094, 0.011] 395 µs	×**0.48** [0.48, 0.49] 8.63 µs	not measured
N=3, R=2	3.54	×**0.14** [0.14, 0.19] 25.2 µs	×**0.43** [0.42, 0.58] 8.30 µs	×**0.096** [0.093, 0.097] 37.4 µs
N=3, R=3	4.09	×**0.062** [0.061, 0.062] 66.2 µs	×**0.38** [0.36, 0.39] 10.7 µs	×**0.066** [0.066, 0.067] 61.7 µs
N=3, R=6	8.28	×**0.0030** [0.0030, 0.0030] 2760 µs	×**0.16** [0.16, 0.16] 51.1 µs	not measured

N=4, R=4	31.6	×**0.060** [0.059, 0.061] 530 µs	×**0.63** [0.62, 0.64] 50.2 µs	×**0.14** [0.13, 0.14] 233 µs
N=4, R=6	48.9	×**0.0039** [0.0039, 0.0040] 12622 µs	×**0.18** [0.18, 0.19] 266 µs	×**0.035** [0.033, 0.035] 1404 µs
N=5, R=5	119	×**0.014** [0.013, 0.014] 8817 µs	×**0.39** [0.38, 0.40] 302 µs	×**0.17** [0.16, 0.17] 711 µs
N=5, R=6	155	×**0.0033** [0.0032, 0.0035] 47123 µs	×**0.15** [0.13, 0.15] 1065 µs	×**0.090** [0.087, 0.090] 1723 µs
N=6, R=6	419	×**0.0020** [0.0020, 0.0021] 205858 µs	×**0.28** [0.27, 0.28] 1522 µs	×**0.31** [0.30, 0.32] 1364 µs
N=7, R=7	4244	skipped (too slow at this batch)	×**0.89** [0.88, 0.90] 4795 µs	×**1.24** [1.20, 1.29] 3439 µs
N=1, R=6 (dot2)	3.14	×**0.090** [0.086, 0.090] 35.0 µs	not measured	not measured
N=2, R=2 (dot2)	1.82	×**0.057** [0.057, 0.057] 31.9 µs	not measured	not measured
N=3, R=4 (dot2)	8.73	×**0.016** [0.016, 0.016] 559 µs	not measured	not measured
N=4, R=6 (dot2)	38.5	×**0.0025** [0.0025, 0.0026] 15138 µs	not measured	not measured
N=4, R=6 (dot3)	62.7	×**0.0033** [0.0033, 0.0034] 18772 µs	not measured	not measured
N=3, R=2 (h1)	3.88	×**0.046** [0.046, 0.047] 83.7 µs	not measured	not measured

1.1 Original IREGI against COLLIER

Reference-only comparison on the shared undotted rows, as absolute times and outcomes from the same capture (no ratio is formed here).

Table 3: Original IREGI (its accepted mode) and COLLIER, µs per point, and their lane outcomes

Integral	IREGI, 1 point	COLLIER, 1 point	IREGI, 32 points	COLLIER, 32 points
N=1, R=0	0.78 IBP	0.39	0.79 IBP	0.42
N=1, R=6	23.4 IBP	4.66	23.4 IBP	4.75
N=2, R=1	3.14 IBP	1.12	3.04 IBP	1.13
N=2, R=6	386 IBP	8.55	395 IBP	8.63
N=3, R=2	24.8 IBP	8.27	25.2 IBP	8.30
N=3, R=3	67.2 IBP	10.7	66.2 IBP	10.7
N=3, R=6	2702 IBP	50.6	2760 IBP	51.1
N=4, R=4	533 IBP	50.4	530 IBP	50.2
N=4, R=6	12484 IBP	265	12622 IBP	266
N=5, R=5	8724 IBP	300	8817 IBP	302
N=5, R=6	46123 IBP	1069	47123 IBP	1065
N=6, R=6	202304 IBP	1952 (unstable 1)	205858 IBP	1522 (success 31, unstable 1)
N=7, R=7	3711318 IBP	5079 (unstable 1)	skipped (too slow at this batch)	4795 (success 31, unstable 1)

2 Amplitudes

Every integral of a one-loop amplitude in one call, 32 phase-space points per call. TIRust shares sub-results between the integrals of a call (results bit-identical to separate evaluation); COLLIER caches them. Each TIRust lane keeps its loop-rotation test and precision rescue.

Table 4: Time per amplitude call and same-capture ratios (ms per call; paired block medians). Shared: one call for all integrals (TIRust's subset store, COLLIER's cache); separate: each integral on its own (no store, no cache)

Amplitude	TIRust shared	vs COLLIER, cache	TIRust separate	vs COLLIER, no cache	TIRust lanes
gg -> gggg, top-loop hexagon, rank 6 (57 integrals)	81.2	×**1.02** [0.98, 1.10] 79.3 ms	158.5	×**0.68** [0.65, 0.69] 235 ms	1824 successful, 0 best effort; shared = separate, bit for bit
gg -> Hgg, top-loop pentagon (g g g H g), rank 5 (26 integrals)	13.6	×**0.78** [0.76, 0.80] 17.7 ms	17.1	×**0.59** [0.57, 0.60] 29.0 ms	832 successful, 0 best effort; shared = separate, bit for bit

3 Accuracy

E/A against the high-precision reference (E/A: dimensionally rescaled whole-array max discrepancy over all Cartesian moments and four Laurent slots (weights mu^(rank+4-2N)) against quadruple-precision original IREGI.) on the stability-study points; NinjaRS on the contracted dense random numerator. "Flagged" counts lanes the code itself marks (TIRust: lanes that used the precision rescue).

Table 5: Per-point E/A: median, 99th percentile and maximum, points worse than 10^{-3} and points without an array

Integral	Code	median	p99	max	> 10^{-3}	no array	flagged
gg -> HH, top-loop box, rank 4 (10000 points)	tirust	1.4e−15	5.2e−12	6.7e−11	0	0	122
	tirust-binary64	1.5e−15	5.3e−12	6.7e−11	0	0	2
	tirust-binary64-r20	1.5e−15	5.5e−11	7.9e−8	0	0	0
	iregi-ibp	2.2e−15	4.7e−11	4.4e−2	2	0	0
	iregi-pv	2.2e−15	6.7e−11	6.8e−2	3	0	0
	collier-coli	1.1e−15	9.1e−12	6.2e−11	0	0	335
	collier-dd	2.0e−15	2.1e−13	8.5e−13	0	0	93
	ninjars (contracted)	3.3e−15	5.3e−11	3.0e−4	0	—	—
gg -> HH kinematics, top-loop box, rank 6 (10000 points)	tirust	1.5e−15	7.5e−12	5.7e−11	0	0	348
	tirust-binary64	1.5e−15	8.0e−12	2.7e−10	0	0	4
	tirust-binary64-r20	1.5e−15	1.1e−9	1.6e−7	0	0	0
	iregi-ibp	2.2e−15	1.2e−8	1.7e2	14	0	0
	iregi-pv	1.4e−1	5.7e0	3.5e2	10000	0	0
	collier-coli	1.7e−15	2.2e−11	2.5e−10	0	0	1499
	collier-dd	2.1e−15	7.8e−13	1.0e−10	0	0	538
	ninjars (contracted)	4.0e−15	1.3e−9	9.0e−3	1	—	—
gg -> Hgg, top-loop pentagon (g g g H g), rank 5 (5000 points)	tirust	3.4e−15	1.2e−11	5.0e−11	0	0	43
	tirust-binary64	3.3e−15	8.5e−12	4.8e−11	0	0	17
	tirust-binary64-r20	3.3e−15	1.2e−11	2.2e−10	0	0	0
	iregi-ibp	2.3e−12	5.0e3	4.2e22	268	0	0
	iregi-pv	6.9e−15	4.4e−9	1.8e−1	2	0	0
	collier-coli	2.9e−15	9.8e−12	1.5e−10	0	0	77
	collier-dd	8.8e−15	1.3e−12	1.3e−9	0	0	473
	ninjars (contracted)	1.7e−14	9.5e−11	1.5e−4	0	—	—
gg -> gggg, top-loop hexagon, rank 6 (2000 points)	tirust	5.9e−14	2.2e−11	5.0e−11	0	0	15
	tirust-binary64	2.6e−14	9.7e−12	3.1e−10	0	0	57
	tirust-binary64-r20	2.8e−14	1.0e−10	3.1e−9	0	0	0
	iregi-ibp	1.2e65	2.3e76	4.1e80	1993	0	0
	iregi-pv	9.9e−13	2.7e−6	2.8e3	6	0	0
	collier-coli	2.1e−14	2.0e−11	2.1e−10	0	0	210
	collier-dd	2.6e−14	2.0e−11	3.4e−10	0	0	985
	ninjars (contracted)	1.3e−13	3.4e−6	6.4e3	7	—	—
gg -> ggggg, top-loop heptagon, rank 7 (500 points)	tirust	1.4e−13	1.9e−11	3.4e−11	0	0	32
	tirust-binary64	8.5e−14	2.6e−11	4.3e−10	0	0	27
	tirust-binary64-r20	8.7e−14	2.3e−10	2.9e−9	0	0	0
	iregi-ibp	7.8e85	1.4e95	1.3e97	495	0	0
	iregi-pv	1.2e2	1.6e5	1.4e6	495	0	0
	collier-coli	8.3e−14	5.0e−11	5.0e−10	0	0	160
	collier-dd	9.8e−14	2.5e−10	5.9e−10	0	0	409
	ninjars (contracted)	4.9e−13	3.0e−4	2.6e0	3	—	—

4 Provenance

- Timing capture: 2026-11-06T15:30:35Z to 2026-11-06T15:46:17Z on CPU 141 (AMD EPYC 9754 128-Core Processor), load 198.64 202.85 191.20 at the start; 11 rounds at batch 1 and 7 at batch 32. Amplitude capture: 2026-11-06T15:46:27Z on CPU 156, 7 rounds.
- TIRust `a0c4884cc5b47e5e6e6c82bc3f1de7719cebdec1` (clean tree for all three builds) binaries: timing `d5862ec9a7570739`, amplitude `ceaa2303799ccff3`, stability `2cc9ddc5790595f1` (SHA-256 prefixes), all with `-C target-feature=+fma`.
- References (SHA-256 prefixes, flags): iregi `f63159d71d60ddb1` (-O3 -std=legacy -fallow-argument-mismatch), collier `29baa9928840c77e` (-O3 -std=legacy -fallow-argument-mismatch -Wtabs), ninjars `22d1353dcbb816f8`.
- Capture summaries (SHA-256): timing `37d746bd769b7f40`, amplitude `2fcd2f7537ef3a1a`, stability `b0b137b17e3f5cef` archived with the build records at the evidence release.